

Stat516hw2

STAT 516 hw2

Several statistics students recorded their heights and the lengths of their index fingers. The measurements, in centimeters and millimeters, respectively, are read into R using the code below (copy and paste this into your .qmd file):

```
link <- url("https://gregorkb.github.io/data/height_length_index_pinky_shoesize_combined.csv")
hl <- read.table(link, sep=",", header=T)
head(hl)
```

	ft	in.	ind	pnk	shoe	shmw	class
1	5	4	75	65	8.0	w	STAT_516_sp_2026
2	5	4	70	56	7.0	w	STAT_516_sp_2026
3	5	8	70	52	10.0	w	STAT_516_sp_2026
4	5	5	68	62	7.5	w	STAT_516_sp_2026
5	5	6	71	55	9.5	m	STAT_516_sp_2026
6	6	4	78	65	13.0	m	STAT_516_sp_2026

```
Y <- (hl$ft*12 + hl$in.)*2.54
X <- hl$ind
n <- length(Y)
```

It is of interest to use the simple linear regression model to predict the height of a person based on the length of his or her index finger.

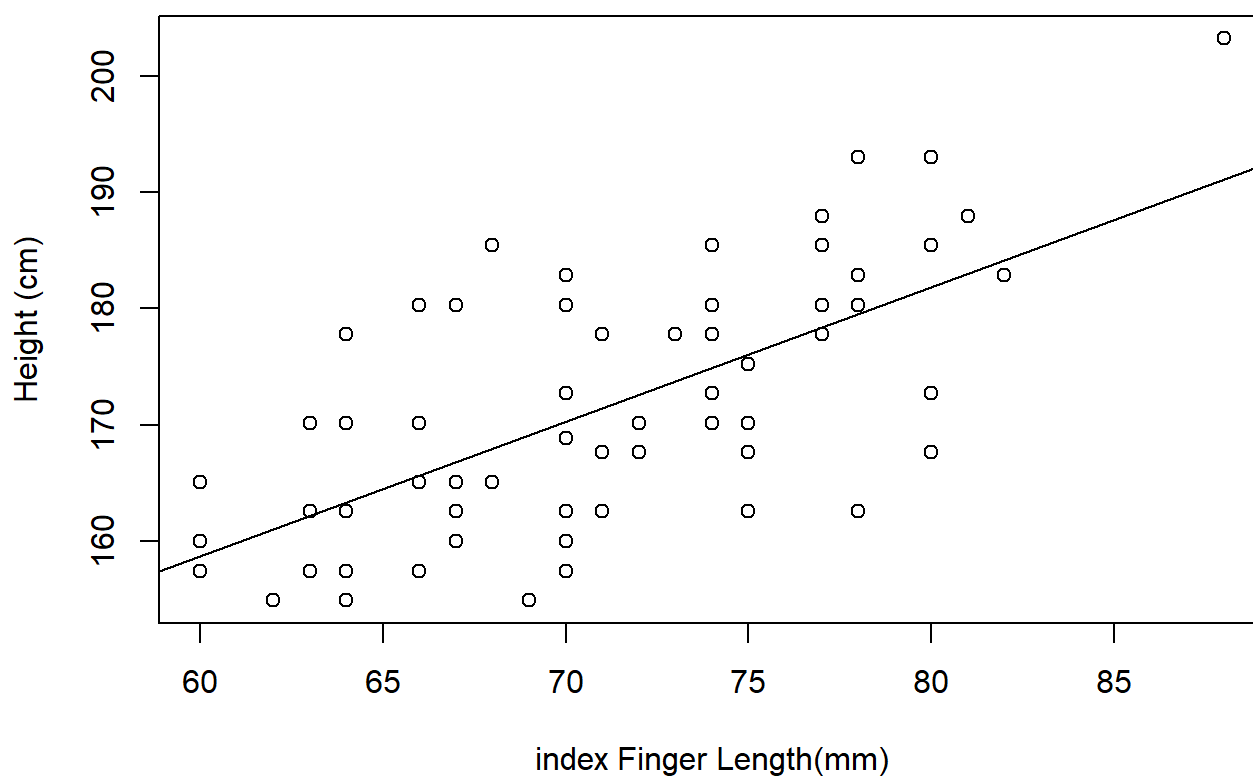
```
fit <- lm(Y ~ X)
```

1)

Make a scatterplot of the heights versus the index finger lengths with the least-squares line overlaid. Report the intercept and slope of the least-squares line as well as the value of Pearson's correlation coefficient.

```
plot(X, Y, xlab = "index Finger Length(mm)", ylab = "Height (cm)", main = "Height vs. Index Finger")
abline(fit)
```

Height vs. Index Finger Length



```
coef(fit)
```

(Intercept)	X
89.36045	1.15670

```
cor(X, Y)
```

```
[1] 0.6644912
```

The intercept of the least squares line is 89.36 and it has a slope of 1.157. the pearson correlation coefficient is 0.6645

2)

Based on the fitted simple linear regression model, to what difference in height does an additional millimeter of index finger length correspond?

```
coef(fit)[2]
```

X
1.1567

based on the linear regression model a additional 1mm of length on the index finger equals an estimated increase of 1.1567cm in height.

3)

Give an estimate of the error term variance.

```
summary(fit)$sigma^2
```

```
[1] 67.66183
```

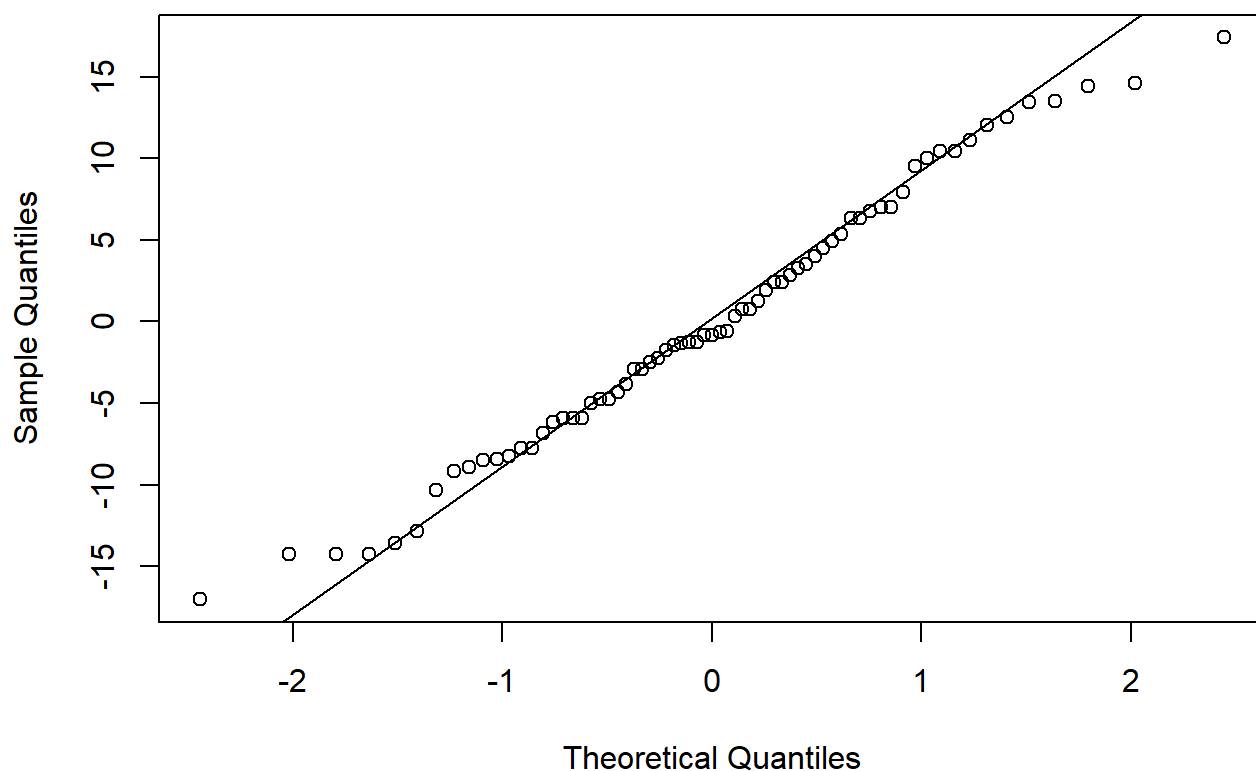
The estimated error term variance is 67.66 cm²

4)

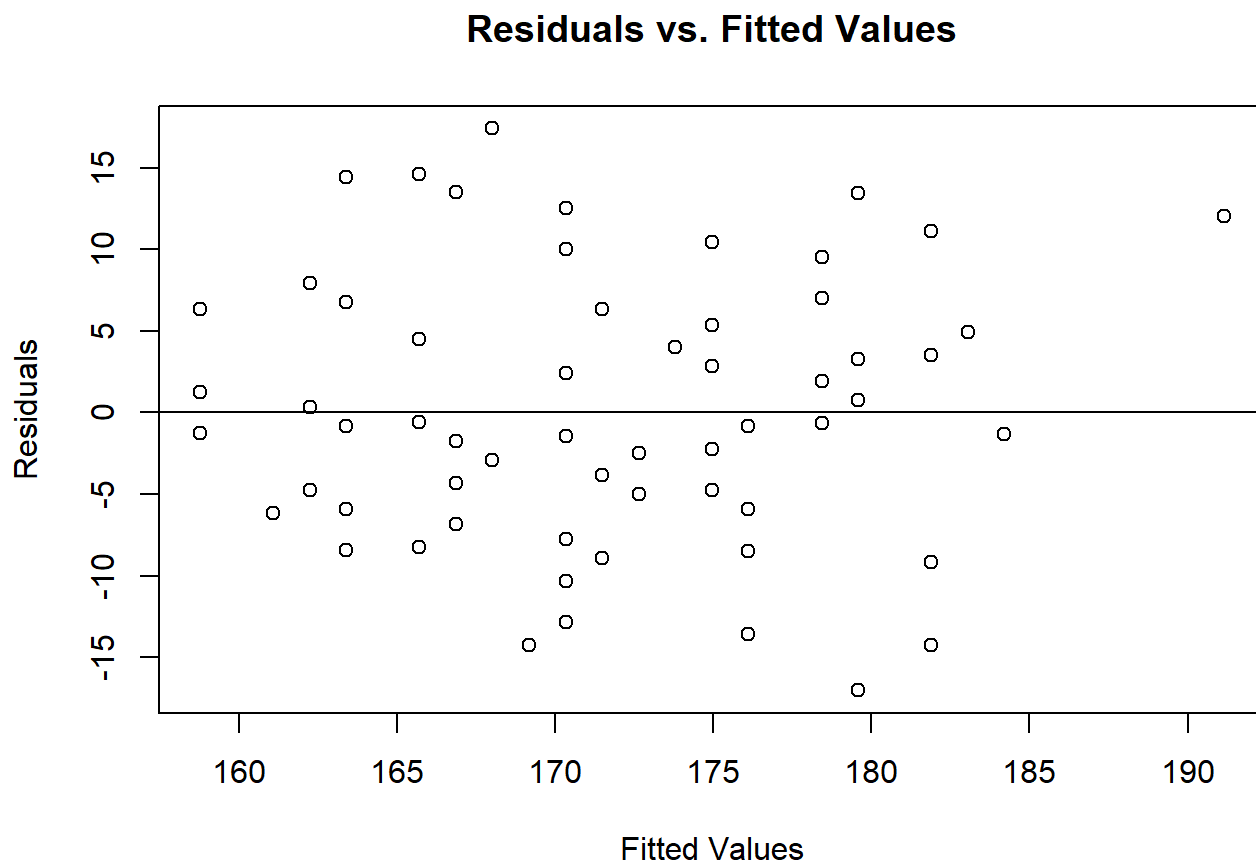
Make a normal quantile-quantile plot of the residuals as well as a residuals versus fitted values plot. Then carefully explain whether you believe the assumptions of the simple linear regression model are satisfied for these data.

```
qqnorm(residuals(fit))  
qqline(residuals(fit))
```

Normal Q-Q Plot



```
plot(fitted(fit), residuals(fit), xlab = "Fitted Values", ylab = "Residuals", main = "Residuals vs  
abline(h = 0)
```



the QQ plot is approximately linear and the residuals are randomly scattered around with no clear pattern. The assumptions of the linear regression model appear to be reasonably satisfied.

5)

Give a 99% confidence interval for the slope parameter .

```
confint(fit, "X", level = 0.99)
```

```
      2.5 %    97.5 %  
X 0.8394881 1.473911
```

The 95% confidence interval is (0.8395, 1.47)

6)

State whether you would reject : versus : at the significance level.

```
summary(fit)
```

Call:

```
lm(formula = Y ~ X)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-17.0230	-5.9329	-0.8292	6.3139	17.4040

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	89.3604	11.3774	7.854	4.41e-11 ***
X	1.1567	0.1589	7.278	4.80e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.226 on 67 degrees of freedom

Multiple R-squared: 0.4415, Adjusted R-squared: 0.4332

F-statistic: 52.97 on 1 and 67 DF, p-value: 4.799e-10

since the p value of 4.8×10^{-10} is less than 0.05 H_0 gets rejected.

7)

Give an estimate of the mean height of persons with index finger length equal to mm. In addition, give a 95% confidence interval for this mean height.

```
new.person <- data.frame(X = 72)

predict(fit, newdata = new.person, interval = "confidence", level = 0.95)
```

	fit	lwr	upr
1	172.6428	170.6545	174.6312

The estimated mean height is 172.64 cm with a 95% confidence interval of (170.65, 174.63) cm.

8)

A hand print is found made by a hand with an index finger length of mm. Give an interval which will contain with probability the height of the person who made the print.

```
new.person <- data.frame(X = 72)

predict(fit, newdata = data.frame(X = 72), interval = "prediction", level = 0.95)
```

	fit	lwr	upr
1	172.6428	156.1043	189.1813

the 95% prediction interval for height is (156.10, 189.18).

9)

Do you think the above interval would be useful in identifying the person who made the print?

No the wouldnt be useful for identifying the person because the range is pretty wide.

10)

Give the value of the coefficient of determination for these data. Interpret the value.

```
summary(fit)$r.squared
```

```
[1] 0.4415485
```

the coefficient of determination is $R^2 = 0.4415$. this means that approximately 44.15% of the variability in height is explained in the linear relationship with finger length

11)

Give the value of the test statistic as well as the value of .

```
summary(fit)
```

Call:

```
lm(formula = Y ~ X)
```

Residuals:

Min	1Q	Median	3Q	Max
-17.0230	-5.9329	-0.8292	6.3139	17.4040

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	89.3604	11.3774	7.854	4.41e-11 ***
X	1.1567	0.1589	7.278	4.80e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.226 on 67 degrees of freedom

Multiple R-squared: 0.4415, Adjusted R-squared: 0.4332

F-statistic: 52.97 on 1 and 67 DF, p-value: 4.799e-10

the value of Ttest is 7.278 and Ftest is 52.97

12)

Give the p-value for testing : versus : based on the value of the test statistic . Interpret your answer.

```
summary(fit)$coefficients
```

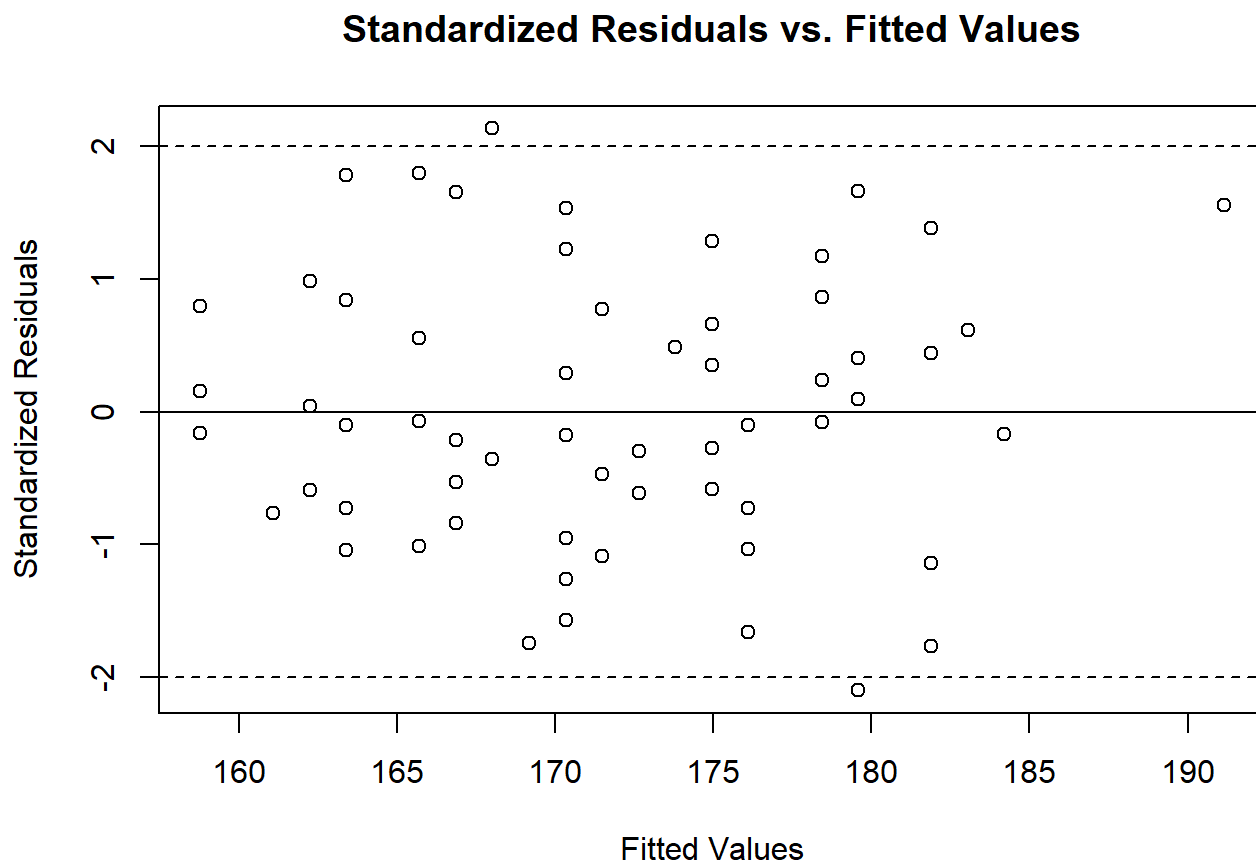
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	89.36045	11.377379	7.854221	4.408067e-11
X	1.15670	0.158923	7.278366	4.799214e-10

the pvalue is 4.799214×10^{-10} . Because the pvalue is less than 0.05 we reject H_0 there is evidence that the slope is not equal to zero.

13)

Comment on whether there are any outliers in the data set. Show a plot to support your answer.

```
plot(fitted(fit), rstandard(fit), xlab = "Fitted Values", ylab = "Standardized Residuals", main =  
abline(h = 0)  
abline(h = 2, lty = 2)  
abline(h = -2, lty = 2)
```



there seems to be one potential outlier because one standardized residual is slightly below -2 and the rest are within the ± 2 range.

14)

Convert the heights to inches (divide by) and fit the regression model again. Report the coefficient of determination as well as the p-value for testing whether the slope coefficient is equal to zero. Compare these to the values you obtained previously.

```
Y.inches <- Y / 2.54

fit.inches <- lm(Y.inches ~ X)

summary(fit.inches)
```

Call:

```
lm(formula = Y.inches ~ X)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.7020	-2.3358	-0.3265	2.4858	6.8520

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	35.18128	4.47928	7.854	4.41e-11 ***
X	0.45539	0.06257	7.278	4.80e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.238 on 67 degrees of freedom

Multiple R-squared: 0.4415, Adjusted R-squared: 0.4332

F-statistic: 52.97 on 1 and 67 DF, p-value: 4.799e-10

after converting the height to inches the coefficient of determination is $R^2 = 0.4415$ and the pvalue is 4.8×10^{-10} . these values are the same so converting to inches does not change it.